

Estimation Sheet

B.Tech CSE 2025–29 · Software Engineering & Project Management · Semester III

Dashboard basis: The project author has developed three dashboards—Student, Faculty and Admin. This document uses the supplied walkthrough to describe them and distinguishes those dashboard details from requirements, integrations or metrics that are proposed or not independently verified.

IN THIS DOCUMENT

1. Inputs supplied in the problem statement • 2. Production effort calculation • 3. Capacity and duration calculation • 4. Revenue projection • 5. Schedule comparison • 6. Assumptions and confidence

1. Inputs supplied in the problem statement

- Recorded content required per course: **12 hours**.
- Combined recording and editing effort: **3 hours per content hour**.
- Editors available: **2 editors**, each with **30 hours/week**.
- Course fee: **₹4,500 per enrolment**.
- Year-one target: **150 enrolments per course**.
- Platform readiness: **10 weeks** (given in the case).

No salary, hosting cost, tax, payment fee, conversion rate or enrolment ramp is supplied; none is added to the revenue projection.

2. Production effort calculation

Per course

Effort per course = content hours × production hours per content hour = $12 \times 3 = 36$ effort-hours per course

For 8 courses

$8 \times 36 = 288$ effort-hours

For 30 courses

$30 \times 36 = 1,080$ effort-hours

The 3-hour factor is treated as combined recording-and-editing effort, as the assignment phrases it. If it means editing only and recording is additional, the estimate would need revision; clarify before baseline approval.

3. Capacity and duration calculation

Weekly capacity

Weekly capacity = editors × hours per editor per week = $2 \times 30 = 60$ effort-hours/week

8-course duration

Duration = total effort ÷ weekly capacity = $288 \div 60 = 4.8$ weeks

30-course duration

= $1,080 \div 60 = 18.0$ weeks

Remaining 22 courses after an 8-course initial batch

Effort = $22 \times 36 = 792$ hours Duration = $792 \div 60 = 13.2$ weeks $4.8 + 13.2 = 18.0$ weeks total

At 60 hours/week, the planned 8-course batch represents 4.8 editor-capacity weeks and the 30-course catalogue represents 18 editor-capacity weeks. This arithmetic does not model course-by-course dependencies, faculty calendars, rework, leave, equipment availability, review time, or the stated faculty limit. The case also says faculty can record only 8 courses in the next three months; therefore, the 4.8-week result should not be interpreted as a promise that 8 fully approved courses will be ready in 4.8 calendar weeks.

4. Revenue projection

Eight courses

Projected enrolments = $8 \times 150 = 1,200$ Revenue = $1,200 \times ₹4,500 = ₹5,400,000 = ₹54,00,000$ (₹54 lakh)

Thirty courses

Projected enrolments = $30 \times 150 = 4,500$ Revenue = $4,500 \times ₹4,500 = ₹20,250,000 = ₹2,02,50,000$ (₹2.025 crore)

These are gross planning projections at the target enrolment count. They are not a forecast of realised collections and exclude refunds, discounts, taxes, payment fees, operating costs and differences in launch timing.

5. Schedule comparison

Scope	Content production estimate	Platform readiness	Planning implication
Initial 8-course batch	4.8 weeks of editor capacity	10 weeks	If content is ready, the platform is the later gate; staged release can be planned for W10.
Full 30-course catalogue	18 weeks of editor capacity	10 weeks	Content takes 8 weeks longer and controls the full-catalogue date; launch all 30 no earlier than W18 under these assumptions.

Full-catalogue content float comparison = $18 - 10 = 8$ weeks of platform float. The platform workstream has eight weeks of total float against the full-catalogue milestone. Content production is the critical path in the simplified two-workstream schedule.

6. Assumptions and confidence

Assumption	Reason / effect
The 3× factor covers all production effort described by the case	Avoids adding an unsupported separate recording estimate. If it does not, effort is understated.
Both editors are available for all 30 hours every week	Gives a theoretical capacity ceiling; leave, meetings and rework make actual duration longer.
Work is divisible and can be scheduled continuously	Real courses may be blocked by faculty, equipment, review or sequencing.
All courses have similar 12-hour content and complexity	Course-specific complexity could make effort uneven.
Revenue target is achieved for every available course	Projection only; no demand evidence was provided.

Confidence: Medium for the arithmetic because the inputs are explicit; low-to-medium for calendar duration because faculty access, course-by-course sequencing, contingency and rework are not quantified. The 18-week result is a capacity-based estimate, **not a promise or contractual delivery date**. Add a contingency only after collecting actual throughput data; do not silently alter the case figures.